WSQ
COURSE

COURSE SLIDES · WSQ

Responsible Generative AI Basics

WSQ Course Code: TGS-2025060472

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v12 · 11 August 2026

COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

1. It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
2. The trainer/administrator displays the digital attendance QR code from the SSG portal.
3. Scan the QR code with your mobile phone camera and submit your attendance.
4. A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer

?

Your Trainer

General Trainer template — to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

CERTIFICATION / CREDENTIALS
PhD; 20+ years of training and industry experience in AI and data.

DELIVERS
WSQ courses on responsible AI, generative AI practices, data analytics and machine learning.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ICE-BREAKER

Let's Know Each Other

1. Your name, organisation and role.
2. Which generative AI tools you already use at work — and what you use them for.
3. One thing about AI that genuinely worries you.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Actively participate — no question is too small.
3. Respect each other's views: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required for funding eligibility.

COURSE ACCREDITATION

Skills Framework

1

TSC Title

Responsible AI and Generative AI Practices

2

TSC Code

ICT-BAS-0055-1.1

3

Assessment

Written Assessment (SAQ) + Case Study — both open book.

4

Structure

3 Learning Units (LU1–LU3), each mapped to a Learning Outcome.

3 LEARNING UNITS

Course Outline

1

LU1 — Ethical Principles of Generative AI

Risks of GenAI Interaction · Ethical Principles · Ethical Decision-Making · Professional Scepticism

2

LU2 — Generative AI Privacy Techniques

Anonymisation & De-identification · Privacy Measures in Practice · Ethical Data Guidelines

3

LU3 — Best Practices of Responsible Generative AI

Responsible AI Principles · Comparing AI Systems · Comparing Ethical Issues

SCHEDULE

Lesson Plan — 1 Day

Morning

- Morning — 9:30am to 12:10pm

– Welcome, ground rules and AM digital attendance

– LU1: Ethical Principles of Generative AI (Activities 1–6)

– Tea break at 11:05am; lunch 12:10–12:55pm

Afternoon & Assessment

- Afternoon — 12:55pm to 7:05pm

– LU1 concludes; LU2: Privacy Techniques (Activities 7–9)

– LU3: Best Practices of Responsible GenAI (Activities 10–14)

– Briefing 5:55pm; Assessment: WA (30 min) + Case Study (30 min), 6:05–7:05pm

- Timing

– 45-minute lunch and two 10-minute tea breaks. Assessment is 1 hour in total.

WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Ethical Reasoning

Apply ethical reasoning to assess generative AI outputs and guide responsible decision-making in AI implementation.

2

LO2 · Privacy Safeguards

Apply privacy safeguards to protect sensitive data used in generative AI through anonymisation and secure storage methods.

3

LO3 · Ethical Comparison

Compare generative AI systems for ethical compliance based on intellectual property, privacy and environmental impact.

BEFORE YOU START

Briefing for Assessment

1. Place phones and other materials under the table or on the floor.
2. No photos or recording of assessment scripts.
3. No discussion during the assessment.
4. Use a black/blue pen for hard-copy assessments.
5. No liquid paper / correction tape.
6. Scripts are collected when time is up.

FINAL ASSESSMENT

Assessment

1. Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book.
2. Case Study (CS) — one continuous scenario with open-ended tasks, 30 minutes, open book.
3. Format: Open Book — slides, Learner Guide and approved materials only.
4. A minimum of 75% attendance (per SSG Digital Attendance record) is required to be eligible for assessment and funding. Learners must be assessed as Competent in every K and A, and complete the TRAQOM survey.
5. An appeal process is available if required.

KNOWLEDGE STATEMENTS · WA-SAQ

What Is Assessed

1

K1 — Written Assessment (SAQ)

Data anonymisation and de-identification techniques

2

K2 — Written Assessment (SAQ)

Ethical considerations and potential risks of generative AI interaction

3

K3 — Written Assessment (SAQ)

Responsible AI principles and best practices for development and deployment, including intellectual property, data privacy and environmental impact considerations

4

K4 — Written Assessment (SAQ)

Ethical principles in AI

What Is Assessed

1. **A1 — Case Study**
   Apply ethical principles in decision-making related to AI

2. **A2 — Case Study**
   Exercise professional scepticism to exercise sound judgement on Gen AI output

3. **A3 — Case Study**
   Apply privacy measures when handling data for AI applications (e.g., implement simple data anonymisation and de-identification techniques, secure data storage using basic encryption)

4. **A4 — Case Study**
   Design guidelines or strategies for ethical AI use, including use of sensitive data in generative AI tools

5. **A5 — Case Study**
   Compare AI systems based on considerations such as compliance with intellectual property, data privacy and environmental impact

6. **A6 — Case Study**
   Compare ethical issues in AI applications

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then the Case Study — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

COURSE PORTAL

Courseware & Assessment on the LMS

Everything for the open-book assessment — slides, Learner Guide and submissions — lives on the LMS.

T

Tertiary Infotech Academy

LMS and TMS for SSG courses

Sign in / OTP

lms-tms.tertiaryinfotech.com

1. **Sign in**
   Log in with your registered email (OTP or password).
2. **Download materials**
   Slides and the Learner Guide from the course page — your open-book references.
3. **Submit answers**
   Upload your Written Assessment and Case Study answers on the LMS.
4. **TRAQOM survey**
   Complete the TRAQOM survey via the QR code shown on the LMS.

https://lms-tms.tertiaryinfotech.com/

LU1

LU1

Ethical Principles of Generative AI

Risks of GenAI Interaction · Ethical Principles · Ethical Decision-Making · Professional Scepticism

Key Concepts — Ethical Principles of Generative AI

1. Generative AI carries hidden costs beyond the subscription fee: environmental load (energy, water, carbon), security exposure, and psychological or societal impact on the people who use it.

2. Deepfakes, voice cloning and face-swapping have made synthetic media cheap and convincing — eroding the assumption that seeing is believing.

3. AI bias is not a single defect: it is introduced through the training data, the model objective, and the deployment context, and it compounds at each stage.

4. Singapore's AI Verify framework defines 11 principles of trustworthy AI, split into process checks and technical tests, and is operationalised through Project Moonshot.

5. Applying ethics in practice means stratifying risk (EU AI Act), keeping a human in the loop for high-stakes decisions, and engineering ethics into the design, not bolting it on.

6. Professional scepticism is the discipline of verifying GenAI output against source evidence — the antidote to automation bias, where people defer to a confident machine.

The Hidden Costs of Generative AI

Every prompt has costs that never appear on the invoice.

Environmental

Training and inference consume electricity, fresh water for cooling and embodied carbon in the hardware. A single large model's training run can emit as much CO_2 as several cars over their lifetime.

Security

Data pasted into a public AI tool may be retained, used for training or exposed. AI also widens the attack surface — prompt injection, model extraction and data leakage are new classes of vulnerability.

Psychological & Societal

Increasing dependence on AI, AI as a social and emotional companion, and the erosion of the human capacity to judge — the 'human filter' degrades the more it is bypassed.

Epistemic

Deepfakes, voice cloning and synthetic media are now cheap and convincing, dissolving the shared assumption that a recording is evidence of anything.

Why GenAI Risk Is Not Hypothetical

The gap between how fast AI is adopted and how fast it is governed is where risk lives.

98%

Accuracy can still mean a life-threatening error in the 2% — accuracy is not safety

3

Layers where bias enters: the training data, the model objective, the deployment context

11

Principles of trustworthy AI defined by Singapore's AI Verify framework

High accuracy on average says nothing about who bears the error. Always ask: who is in the 2%?

The Deconstruction of AI Bias

Bias is not one defect to be patched — it is introduced at three separate stages and compounds.

	How it enters	What it produces
Data bias	Training data over-represents some groups and under-represents others; historical records encode past discrimination.	A model that is systematically less accurate for under-represented groups.
Objective bias	The metric the model optimises (clicks, cost, speed) is a proxy for what we actually care about (welfare, fairness, safety).	A model that optimises the proxy perfectly while defeating the real goal.
Deployment bias	The system is used on a population, or for a purpose, different from the one it was built and validated for.	Confident predictions in a context where the model was never tested.

AI Verify — Singapore's Testing Framework

AI Verify splits the 11 principles by how each can actually be evidenced. aiverifyfoundation.sg

Zone 1 — Process checks only

Transparency · Reproducibility · Safety · Security · Data Governance · Accountability · Human Agency & Oversight · Inclusive Growth & Societal Impact. Evidenced through standardised checklists, documentation and governance frameworks.

Zone 2 — Technical tests + process checks

Fairness · Explainability · Robustness. Evidenced through black-box technical testing on model inputs and outputs, supported by quantitative toolkits.

Why the split matters

You cannot unit-test accountability, and you cannot govern robustness with a policy document alone. The framework integrates technical validation (TEVV) directly with process governance.

Project Moonshot

The open-source companion toolkit that operationalises the testing — benchmarking, red-teaming and baseline testing for LLM applications. github.com/aiverify-foundation/moonshot

GenAI Risk Diagnostic Matrix

Each GenAI risk category has a distinct testing target — you cannot test them all the same way.

	Output testing target	Component testing target
Hallucination & inaccuracy	Domain knowledge; out-of-domain handling	Retrieval-Augmented Generation (RAG)
Bias in decision making	Parity metrics; counterfactual checks	Model, system prompt, input filters
Undesirable content	Content type; elicitation rate; helpfulness	Input & output filters; system prompt
Data leakage	Data type leaked; ease of elicitation	Input & output filters; system prompt
Adversarial prompts	Direct vs indirect prompt injection	Input & output filters; system prompt

The EU AI Act Compliance Waterfall

The EU AI Act regulates by the risk a system poses, not by the technology it uses.

1. Unacceptable risk — prohibited outright (social scoring, manipulative systems)
2. High risk — conformity assessment, risk management, human oversight, logging
3. Limited risk — transparency obligations (users must know they are dealing with AI)
4. Minimal risk — no additional obligation; voluntary codes of conduct

Classify the system FIRST — every downstream obligation follows from the risk tier.

Three Pillars of Algorithmic Risk

Transparency

Neither the affected person nor often the operator can see why the system produced its output — so the decision cannot be contested or corrected.

Algorithmic bias

Systematically unfair outcomes for particular groups, arising from data, objective or deployment context.

Data privacy

Personal data is used for training or inference without a lawful basis, adequate safeguards or the individual's awareness.

The governing response

National AI policy — the EU AI Act, NIST AI 600-1, Canada's AIDA and Singapore's AI Verify / Model AI Governance Framework — each address these three pillars with different instruments.

LU1 · T3 · THE NON-NEGOTIABLE CONTROL

The ultimate safeguard requires a human in the loop.

Automation is acceptable where the cost of an error is low and reversible. Where it is high-stakes and irreversible, a qualified human must remain accountable for the decision — not merely present to rubber-stamp it.

Automation Bias — The Vulnerability in the Human Filter

Human oversight fails predictably when people defer to a confident machine.

What it is

The tendency to over-trust an automated system — accepting its output without the scrutiny you would apply to a human colleague's work.

Why it happens

Fluency reads as competence. AI output is grammatical, confident and well-formatted, and those cues are the ones humans use to judge reliability.

How it compounds

Time pressure, high volume and a good track record all reduce checking — so the review becomes a formality exactly as the system's errors become rarer and harder to spot.

The counter-measure

Structured verification: check the claim against the source, not against your impression of the output. Verify names, numbers, citations and dates independently every time.

WHAT YOU'LL DO

In-Class Activities — Ethical Principles of Generative AI

Activities 1–2

- AI Ethical Dilemma Simulator
- AI Ethical Principles Quiz

Activities 3–4

- Prompt Injection Playground
- Ethical Decision-Making Framework

Activities 5–6

- AI Output Scepticism Checker
- The Cognitive Threat Matrix

LU1 · ACTIVITY 1

AI Ethical Dilemma Simulator

ACTIVITY 1

Work through six real-world AI dilemmas — from AI-generated hospital discharge summaries to resume-screening bias — choosing the most defensible course of action in each and reading the consequences of the choice you made.

WORKFLOW

1. Read the dilemma
2. Choose a response
3. Study the analysis
4. Repeat across 6 domains
5. Record your score

YOU'LL PRODUCE

A completed 6-scenario run with your score out of 6 and notes on the stakeholders and ethical principles at stake in each dilemma

Duration
10 minutes

LU1 · ACTIVITY 2

AI Ethical Principles Quiz

ACTIVITY 2

Test your grasp of the principles that underpin trustworthy AI — fairness, transparency, accountability, privacy, beneficence and human agency — across ten applied questions.

WORKFLOW

1. Answer 10 questions
2. Read each explanation
3. Track correct / incorrect
4. Review your score
5. Retake the weak areas

YOU'LL PRODUCE

A completed 10-question quiz with your score and the live Correct / Incorrect counters

Duration

10 minutes

Prompt Injection Playground

ACTIVITY 3

Attack three simulated AI assistants — an undefended customer-service bot, a banking assistant with basic defences and a medical triage AI with strong defences — and see which injections get through each layer of protection.

WORKFLOW

1. Learn the attack types
2. Attack Level 1
3. Escalate to Level 2 & 3
4. Read each verdict
5. Record the scoreboard

YOU'LL PRODUCE

An Attack Scoreboard showing your total attempts, successful injections, success rate and which of the three scenarios you breached

Duration
10 minutes

Ethical Decision-Making Framework

ACTIVITY 4

Take one contested AI deployment through a six-stage structured ethical analysis — scenario, stakeholders, risks, principles, alternatives, decision — and produce a defensible, documented recommendation rather than a gut reaction.

WORKFLOW

1. Define the scenario
2. Map stakeholders
3. Weigh risks & benefits
4. Apply the principles
5. Decide & plan monitoring

YOU'LL PRODUCE

A printed 'Ethical Decision Analysis Summary' report with all eight sections completed

Duration

15 minutes

AI Output Scepticism Checker

ACTIVITY 5

Evaluate confident, fluent, and quietly wrong AI-generated text against a 16-point scepticism checklist, and score how far it can actually be trusted.

WORKFLOW

1. Load AI-generated text
2. Work the 16 checks
3. Set each severity
4. Calculate the score
5. Read the recommendations

YOU'LL PRODUCE

A scepticism score out of 100 with a risk verdict and the recommendations list for the text you assessed

Duration
10 minutes

The Cognitive Threat Matrix

ACTIVITY 6

Pair a human cognitive bias with the AI failure mode it amplifies — automation bias with hallucination, confirmation bias with skewed data — and generate a structured risk analysis of a scenario from your own workplace.

WORKFLOW

1. Enter your API key
2. Pick a threat pairing
3. Load or write a scenario
4. Generate the analysis
5. Extract the mitigations

YOU'LL PRODUCE

A five-part AI risk analysis covering scenario overview, the bias at play, risk escalation, real-world impact and mitigation strategies

Duration
15 minutes

Activity Links — Ethical Principles of Generative AI

1. **Activity 1: AI Ethical Dilemma Simulator**
https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu1-ethical-principles/ethical-dilemma/

2. **Activity 2: AI Ethical Principles Quiz**
https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu1-ethical-principles/principles-quiz/

3. **Activity 3: Prompt Injection Playground**
https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu1-ethical-principles/prompt-injection/

4. **Activity 4: Ethical Decision-Making Framework**
https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu1-ethical-principles/decision-framework/

5. **Activity 5: AI Output Scepticism Checker**
https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu1-ethical-principles/skepticism-checker/

6. **Activity 6: The Cognitive Threat Matrix**
https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu1-ethical-principles/cognitive-threat-matrix/ · Needs a Gemini API key

LU1 RECAP

Recap — Ethical Principles of Generative AI

1

AI Ethical Dilemma Simulator

You can now: Recognise the ethical considerations and potential risks that arise when interacting with generative AI (K2).

2

AI Ethical Principles Quiz

You can now: Identify and apply the core ethical principles in AI (K4).

3

Prompt Injection Playground

You can now: Explain how prompt injection threatens AI safety and how layered defences reduce the risk (K2).

4

Ethical Decision-Making Framework

You can now: Apply ethical principles systematically in decision-making related to AI (A1).

5

AI Output Scepticism Checker

You can now: Exercise professional scepticism to exercise sound judgement on generative AI output (A2).

6

The Cognitive Threat Matrix

You can now: Analyse how human cognitive bias combines with generative AI failure modes to escalate risk (K2, A2).

LU2

LU2

Generative AI Privacy Techniques

Anonymisation & De-identification · Privacy Measures in Practice · Ethical Data Guidelines

Key Concepts — Generative AI Privacy Techniques

1. Anonymisation irreversibly severs the link to an individual; pseudonymisation only replaces identifiers and remains personal data under the PDPA and GDPR.

2. K-anonymity guarantees an individual cannot be distinguished from at least k-1 others by generalising or suppressing quasi-identifiers.

3. Release-based privacy fails when an attacker holds external auxiliary data — the re-identification attack that motivated stronger, computation-based guarantees.

4. The three layers of AI data security are the pipeline (encryption in transit and at rest), the model (training and memorisation controls) and the output (filtering).

5. Format-preserving encryption and data-shifting keep a field usable by downstream systems while removing its identifying power.

6. An ethical AI data policy classifies data as prohibited, restricted or permitted for AI input, and pairs each class with a mandated safeguard.

The Legal Baseline — Anonymisation vs Pseudonymisation

Getting this wrong is the single most common privacy compliance error in AI projects.

	Pseudonymisation	Anonymisation
What it does	Replaces identifiers with a token or key; the link to the individual is retained somewhere.	Irreversibly severs the link — no key exists that can restore identity.
Reversible?	Yes — by anyone holding the mapping key.	No — by design, not by policy.
Still personal data?	YES. Remains fully in scope of the PDPA and GDPR, with all obligations intact.	No. Out of scope, because there is no longer an identifiable individual.
Typical use	Internal analytics where re-identification must remain possible for legitimate operational reasons.	Publishing, external sharing, or training data where re-identification must be impossible.

The Mechanics of K-Anonymity

Guaranteeing an individual cannot be distinguished from at least k-1 others.

1. Identify direct identifiers (name, NRIC, email) and remove them
2. Identify quasi-identifiers (age, postcode, gender) that combine to re-identify
3. Generalise values — exact age becomes an age band, postcode loses digits
4. Suppress records that remain unique even after generalisation
5. Verify every combination of quasi-identifiers appears at least k times

Its limit: k-anonymity fails when the attacker holds external auxiliary data, or when every record in a group shares the same sensitive value.

The Three Layers of AI Data Security

Protecting data in an AI system means protecting it at rest, in computation and at output.

Layer 1 — The pipeline

Encryption in transit and at rest; access control; format-preserving encryption so a field stays usable by downstream systems while losing its identifying power; data shifting and masking.

Layer 2 — The model

Controls on what enters training; the training paradox of balancing data utility against data privacy; differential privacy and DP-SGD to bound what any one record can contribute.

Layer 3 — The output

Output filtering to catch memorised training data; guarding against LLM memorisation, where a model reproduces verbatim sensitive strings it saw during training.

The governance wrapper

Know your tools · establish policy · lock down inputs · understand external threats · update training · review vendors · plan for incidents.

7-Step Governance Checklist for AI Adoption

This checklist is directly assessable — you should be able to reproduce all seven steps.

1 · Know your tools

Inventory every AI tool actually in use, including the shadow IT that staff adopted without approval.

2 · Establish policy

Classify data as prohibited, restricted or permitted for AI, and publish it.

3 · Lock down inputs

Anonymise or mask data before anything reaches a model.

4 · Understand external threats

Prompt injection, data exfiltration, model extraction and vendor breach.

5 · Update training

Staff must know the limits of the tools they are given.

6 · Review vendors

Retention, training use, sub-processors, jurisdiction and incident history.

7 · Plan for incidents

Detection, reporting deadline, assessment, notification and remediation.

Choosing the Right Protection

Each technique buys a different guarantee at a different cost to utility.

	Protects against	Cost
Masking / redaction	Casual inspection of directly identifying fields.	Cheap and fast, but destroys the field's analytical value entirely.
K-anonymity	Singling out an individual from quasi-identifiers.	Moderate loss of granularity; fails against auxiliary-data attacks.
Differential privacy	Inferring whether any individual was in the dataset at all.	Adds calibrated noise — a formal guarantee paid for in accuracy.
Homomorphic encryption	Exposure during computation — data never decrypted to be processed.	Strongest guarantee, but computationally very slow.
Synthetic data	Direct exposure of any real record.	Useful when masking destroys utility; fidelity and residual leakage must be validated.

Designing an Ethical AI Data Policy

01

Classify

Sort every data category into PROHIBITED (never enters an AI tool), RESTRICTED (only with safeguards) and PERMITTED (freely usable).

02

Comply

Name the regimes that bind you — PDPA as the Singapore baseline, plus GDPR, HIPAA or the EU AI Act where applicable.

03

Safeguard

Mandate the specific controls: anonymisation before input, encryption, role-based access, audit logging and human review of outputs.

04

Respond

Define detection, a reporting deadline, severity assessment, notification and a post-incident review that updates the policy.

Trust is engineered, not accidental — a policy nobody can follow protects nobody.

WHAT YOU'LL DO

In-Class Activities — Generative AI Privacy Techniques

Activities 7–8

- Data Anonymisation Techniques
- Python Data Anonymisation Pipeline

Activities 9–9

- Ethical AI Data Policy Generator

Data Anonymisation Techniques

ACTIVITY 7

Run one set of sensitive records through four different anonymisation techniques side by side, and see why only some of them actually remove the re-identification risk.

WORKFLOW

1. Enter your API key
2. Generate sample records
3. Apply all 4 techniques
4. Compare the techniques
5. Compare the reviewers

YOU'LL PRODUCE

A side-by-side comparison table showing the same record under all four techniques, including three different reviewers' subjective judgements

Duration
15 minutes

Python Data Anonymisation Pipeline

ACTIVITY 8

Build a seven-stage anonymisation pipeline over a raw dataset — pseudo-IDs, text shifting, masking, generalisation and column dropping — then export both the anonymised data and the Python code that produced it.

WORKFLOW

1. Load a raw dataset
2. Select the techniques
3. Run the pipeline
4. Inspect the output
5. Export CSV + code

YOU'LL PRODUCE

An exported anonymised CSV plus the generated Python script implementing your chosen techniques

Duration

15 minutes

LU2 · ACTIVITY 9

Ethical AI Data Policy Generator

ACTIVITY 9

Produce a complete, organisation-specific AI data-handling policy through a guided five-step wizard — the document that tells your colleagues what they may and may not paste into an AI tool.

WORKFLOW

1. Profile the organisation
2. Classify the data
3. Select the regulations
4. Choose the safeguards
5. Generate & print

YOU'LL PRODUCE

A printed seven-section 'AI Data Handling Policy' tailored to your organisation, sector and risk tolerance

Duration

15 minutes

Activity Links — Generative AI Privacy Techniques

1. **Activity 7: Data Anonymisation Techniques**
   https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu2-privacy/dataprivacy/ · Needs a Gemini API key

2. **Activity 8: Python Data Anonymisation Pipeline**
   https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu2-privacy/anonymizer-python/

3. **Activity 9: Ethical AI Data Policy Generator**
   https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu2-privacy/privacy-policy/

LU2 RECAP

Recap — Generative AI Privacy Techniques

1

Data Anonymisation Techniques

You can now: Distinguish the anonymisation spectrum — de-identification, pseudonymisation, objective and subjective anonymisation (K1).

2

Python Data Anonymisation Pipeline

You can now: Apply privacy measures when handling data for AI applications by building an anonymisation pipeline (A3).

3

Ethical AI Data Policy Generator

You can now: Design guidelines or strategies for ethical AI use, including use of sensitive data in generative AI tools (A4).

LU3

LU3

Best Practices of Responsible Generative AI

Responsible AI Principles · Comparing AI Systems · Comparing Ethical Issues

Key Concepts — Best Practices of Responsible Generative AI

1. The EU AI Act regulates by risk tier — unacceptable, high, limited and minimal — with obligations that scale to the harm the system can cause.

2. ISO/IEC 42005:2025 gives an internal, non-certifiable structure for assessing an AI system's impact on individuals, groups and society across its lifecycle.

3. Differential privacy is a mathematical guarantee: the output does not reveal whether any single individual's record was in the dataset, tuned by the epsilon budget.

4. Comparing AI systems responsibly means scoring them on intellectual property provenance, data privacy posture and environmental cost — not accuracy alone.

5. 'Green AI' routes appropriate tasks to small language models, cutting energy and cost against the 'Red AI' default of always using the largest frontier model.

6. Explainability trades off against raw accuracy; XAI frameworks such as SHAP and LIME recover interpretability where the decision affects people's lives.

The Three Pillars of Safe AI

Intellectual property

Was the training data licensed? Who owns the output? Getty Images v Stability AI put provenance of training data at the centre of the question.

Data privacy

What personal data entered the model, under what lawful basis, and can it be extracted again through memorisation?

Environmental impact

Energy, water and carbon per training run and per inference — the cost that scales with every user you add.

The unifying question

Responsible comparison scores systems on all three, not on benchmark accuracy alone. A more accurate model with unlicensed training data and no privacy posture is not the better choice.

ISO/IEC 42005 — The Assessment Lifecycle

ISO/IEC 42005:2025 — internal guidance for assessing an AI system's impact on individuals, groups and society. Not a certification standard; you do not need an external auditor.

1. Define scope, purpose and the context of use
2. Identify affected individuals, groups and society
3. Assess potential harms and benefits across the lifecycle
4. Determine controls and mitigations for each identified harm
5. Document the assessment and its residual risk
6. Review and re-assess when the system or its context changes

Use it internally to bring structure to how consequences are evaluated — from planning and design through deployment and monitoring.

Differential Privacy — The Epsilon Budget

Epsilon (ε) quantifies privacy loss: lower means stronger privacy and noisier answers.

$\varepsilon \leq 1$

Strong privacy — high noise, suitable for publishing sensitive statistics

$\varepsilon \approx 1–10$

Moderate — the practical working range for most deployed systems

$\varepsilon > 10$

Weak — the formal guarantee becomes close to meaningless

DP-SGD applies this during training: clip each example's gradient to bound its influence, then add calibrated noise — so no single person's data can be recovered from the model.

Comparison of AI Regulations

There is no single global AI regime — comparing systems means comparing against several.

	Approach	Binding force
EU AI Act	Risk-tiered: prohibited, high, limited and minimal risk, with obligations scaled to the tier.	Binding law with substantial financial penalties.
NIST AI 600-1 (US)	A companion to the AI Risk Management Framework addressing risks specific to generative AI.	Voluntary framework; influential in procurement and practice.
Canada AIDA	Focused on 'high-impact' systems, with obligations on assessment and mitigation.	Proposed legislation.
Singapore AI Verify	Testing framework of 11 principles across process checks and technical tests, with the Model AI Governance Framework.	Voluntary, industry-led; the practical baseline for Singapore organisations.

Red AI vs Green AI

Task alignment — routing work to the smallest model that can do it — is the single largest lever on AI's environmental cost.

	'Red AI' — frontier by default	'Green AI' — right-sized
Model choice	Always the largest available frontier model, regardless of task difficulty.	A small language model (SLM) for well-scoped, repetitive tasks; frontier models reserved for genuinely hard ones.
Energy per query	Orders of magnitude higher; every user added multiplies it.	Dramatically lower, and often runs on local or modest hardware.
Cost & latency	Highest cost per call and slower responses.	Cheaper and faster — the business case usually aligns with the green case.
When justified	Open-ended reasoning, novel synthesis, complex multi-step work.	Classification, extraction, routing, summarisation, structured formatting.

Openness Does Not Equal Transparency

'Open-weights' is a licence claim, not a disclosure of how the system was built.

What is usually disclosed

Model weights, an architecture description, an inference licence and benchmark scores.

What usually is not

The training-data composition and provenance, the data-filtering decisions, the human feedback process and the evaluation failures.

Why it matters for compliance

You cannot assess IP provenance or privacy risk from weights alone — the questions a regulator asks concern the data, not the parameters.

What to ask a vendor

Data provenance and licensing; retention and training use of your inputs; published evaluation results including failures; incident history and escalation path.

The Technical Trade-off — Accuracy vs Interpretability

The more capable the model, the harder it is to explain — and explanation is often a legal requirement.

	Interpretable by design	Black box + post-hoc XAI
Examples	Linear and logistic regression, decision trees, rule lists.	Deep neural networks, gradient-boosted ensembles, large language models.
Explanation	The model IS the explanation — you can read the coefficients or the tree.	Requires SHAP or LIME to approximate why a particular decision was made.
Trade-off	Lower ceiling on accuracy for complex, high-dimensional problems.	Higher accuracy, but the explanation is an approximation, not the mechanism.
Where each fits	Regulated, contestable decisions — credit, hiring, benefits, sentencing.	Perception, language and pattern tasks — with XAI bolted on where decisions affect people.

LIME vs SHAP

	LIME	SHAP
Basis	Fits a simple interpretable model locally around the one prediction being explained.	Shapley values from cooperative game theory — each feature's fair share of the outcome.
Consistency	Stochastic — re-running gives slightly different values each time.	Deterministic and additive — the contributions sum exactly to the prediction.
Speed	Fast; practical for quick, local inspection.	Slower; exact computation is expensive for large models.
Best for	Rapid, intuitive checks on individual predictions during development.	Explanations that must be defended to a regulator, an auditor or the affected person.

The Anatomy of Algorithmic Discrimination

The same failure pattern recurs across documented cases in hiring, credit, health and policing.

Amazon's recruiting tool

Trained on ten years of predominantly male hires; learned to penalise resumes containing signals of being a woman. Withdrawn.

Apple Card

Couples with shared finances received markedly different credit limits; neither the operator nor the issuer could explain the decisions.

IBM Watson for Oncology

Recommended unsafe treatments; trained substantially on synthetic cases rather than real patient outcomes.

Clearview AI

Scraped billions of facial images without consent to build an identification service sold to law enforcement.

LU3 · T3 · THE ULTIMATE OBJECTIVE

Ethical AI is not a roadblock to innovation — it is the prerequisite.

Cost reduction must advance alongside accessibility and quality. A system that is cheaper and faster but unfair, opaque or unsafe has not improved anything that matters.

WHAT YOU'LL DO

In-Class Activities — Best Practices of Responsible Generative AI

Activities 10–11

- AI System Comparison Matrix
- Differential Privacy Explorer

Activities 12–13

- Explainable AI (XAI) Explorer
- The Ethical Paradigms of AI Resource Allocation

Activities 14–14

- AI Ethics Case Study Analyzer

AI System Comparison Matrix

ACTIVITY 10

Score ChatGPT, Claude and Gemini across five responsibility dimensions and plot them on a radar chart — turning 'which AI should we use?' into an evidence-based comparison rather than a brand preference.

WORKFLOW

1. Score ChatGPT
2. Score Claude & Gemini
3. Add a 4th system
4. Update the chart
5. Read the radar & table

YOU'LL PRODUCE

A radar comparison chart and scored comparison table ranking at least three AI systems by overall average

Duration

15 minutes

LU3 · ACTIVITY 11

Differential Privacy Explorer

ACTIVITY 11

Turn the epsilon dial and watch the privacy-utility trade-off happen in front of you — strong privacy makes the answer noisy, weak privacy makes it accurate but leaks.

WORKFLOW

1. Run at epsilon 1.0
2. Compare low vs high
3. Try randomised response
4. Spend the budget
5. Exhaust the budget

YOU'LL PRODUCE

A set of query results at three different epsilon values plus a spent privacy budget showing how many queries a dataset can safely answer

Duration
15 minutes

Explainable AI (XAI) Explorer

ACTIVITY 12

Open the black box on a loan-approval model: adjust an applicant's profile, watch the decision flip, and compare how LIME and SHAP each explain why.

WORKFLOW

1. Load a borderline case
2. Flip the decision
3. Run LIME
4. Run SHAP
5. Compare side by side

YOU'LL PRODUCE

LIME and SHAP explanations for the same borderline applicant, plus the side-by-side comparison of the two methods

Duration
15 minutes

The Ethical Paradigms of AI Resource Allocation

ACTIVITY 13

Allocate a scarce resource — 50 treatments among 120 eligible patients — three times over, once under each ethical paradigm, and confront the fact that all three are defensible and they disagree about who lives.

WORKFLOW

1. Study the 3 paradigms
2. Load the scenario
3. Run all 3 allocations
4. Compare the outcomes
5. Justify your choice

YOU'LL PRODUCE

Three allocation results for the same scenario under the Utilitarian, Egalitarian and Prioritarian paradigms, with your own written justification of which is most appropriate

Duration
15 minutes

AI Ethics Case Study Analyzer

ACTIVITY 14

Analyse documented AI ethics failures — IBM Watson for Oncology, the Apple Card credit investigation, Amazon's recruiting tool, Clearview AI — and propose the remediation that should have been in place.

WORKFLOW

1. Filter to your sector
2. Read the case
3. Identify the issues
4. Propose remediation
5. Find the pattern

YOU'LL PRODUCE

A written analysis of at least one case identifying its ethical issues and your proposed remediation, compared against an AI critique

Duration
15 minutes

Activity Links — Best Practices of Responsible Generative AI

1. **Activity 10: AI System Comparison Matrix**
   https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu3-best-practices/ai-comparison/

2. **Activity 11: Differential Privacy Explorer**
   https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu3-best-practices/differential-privacy/

3. **Activity 12: Explainable AI (XAI) Explorer**
   https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu3-best-practices/xai-explorer/

4. **Activity 13: The Ethical Paradigms of AI Resource Allocation**
   https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu3-best-practices/ai-resource-allocation/ · Needs a Gemini API key

5. **Activity 14: AI Ethics Case Study Analyzer**
   https://tertiarycourses.github.io/TGS-2025060472-Responsible-Generative-AI-Basics/lu3-best-practices/ethics-case-study/ · Needs a Gemini API key

LU3 RECAP

Recap — Best Practices of Responsible Generative AI

1

AI System Comparison Matrix

You can now: Compare AI systems on intellectual property, data privacy and environmental impact (A5).

2

Differential Privacy Explorer

You can now: Explain how differential privacy protects individuals and the privacy-utility trade-off it imposes (K1, K3).

3

Explainable AI (XAI) Explorer

You can now: Compare explainability techniques and the accuracy-interpretability trade-off (K3, A6).

4

The Ethical Paradigms of AI Resource Allocation

You can now: Compare ethical issues in AI applications across competing ethical frameworks (A6).

5

AI Ethics Case Study Analyzer

You can now: Identify ethical issues in real AI deployments and propose remediation (A6, A1).

WRAP-UP

Course Summary & Next Steps

LEARNING OUTCOMES

What You Achieved

1

LO1 · Ethical Reasoning

Assessed generative AI outputs against ethical principles, exercised professional scepticism and made documented, defensible decisions.

2

LO2 · Privacy Safeguards

Applied anonymisation, de-identification and secure-storage measures, and designed an ethical AI data policy.

3

LO3 · Ethical Comparison

Compared AI systems on intellectual property, data privacy and environmental impact, and analysed real ethical failures.

WRAP-UP

Assessment

1. Written Assessment (WA, SAQ) — 30 minutes, covering K1–K4. Case Study (CS) — 30 minutes, covering A1–A6.

2. Open book: slides, Learner Guide and approved materials only.

3. Remember to complete the TRAQOM survey and take the Assessment digital attendance (two separate steps).

4. Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then the Case Study — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

SSG DIGITAL ATTENDANCE

Digital Attendance (Mandatory)

1. It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
2. The trainer/administrator displays the digital attendance QR code from the SSG portal.
3. Scan the QR code with your mobile phone camera and submit your attendance.
4. A minimum of 75% attendance is required to be eligible for assessment and funding.

SEE YOU AT THE NEXT ONE

Thank You!

You can now question a confident machine, protect the data you feed it, and compare AI systems on more than accuracy alone.